Probability with Engineering Applications

ECE 313 • MATH 362

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Preface

These course notes for ECE 313/MATH 362: Probability with Engineering Applications present a coherent, lecture-aligned treatment of probability for engineers. The material develops foundational ideas (events, axioms, counting, conditional probability, Bayes' theorem, independence), discrete and continuous random variables with expectations and inequalities, core distributions (Bernoulli, Binomial, Geometric, Negative Binomial, Poisson, Uniform, Exponential, Gaussian), transformations, random processes (Bernoulli to Poisson), joint distributions with covariance and correlation, and limit theorems (central limit theorem and Gaussian approximation). Estimation and hypothesis testing are treated in both discrete and continuous settings, and reliability analysis includes union-bound approximations and design trade-offs.

Each chapter lists topics to cover and learning objectives, includes a brief KEEN 3Cs framework (curiosity, connections, creating value) to emphasize opportunity recognition, integration of knowledge, and stakeholder impact, and concludes with practice problems organized from recall and understanding to application, analysis, evaluation, and creation. Cross-references connect recurring tools such as conditioning, LOTUS, and inequalities across chapters to reinforce transfer and durable understanding in engineering contexts.

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Chapter 1

Course Introduction and Embracing Uncertainty

Topics to Cover

- What probability means in engineering applications
- Expected value and risk–neutral decisions
- Forecasting and incentives through payoff-based interpretations

Learning Objectives

- Identify actions, outcomes, probabilities, and payoffs in simple decision problems.
- Compute expected value and apply a risk–neutral decision rule.
- Derive and interpret why a payoff rule leads a forecaster to report her true belief.

1.1 Why probability

If a fair die is rolled, we say the probability of rolling a six is 1/6. How do we know? If the die is rolled a million times, about one-sixth of the outcomes are sixes.

If a weather forecaster says "the probability of rain tomorrow is 30%," how should we interpret that? What does "30%" mean? One way to understand probability is to look at the *incentives* under which a forecaster makes a report. If we understand the rules determining her payoffs, then we gain structural clarity about what her reported number represents.

This chapter introduces two examples that motivate these ideas.

1.2 Example A: stock decision (risk–neutral baseline)

Tomorrow's stock movement falls into one of:

$$\text{Very High (V)}, \quad \text{High (H)}, \quad \text{Low (L)}.$$

Let returns be r_V, r_H, r_L, and subjective probabilities be p_V, p_H, p_L.

A risk–neutral decision maker buys the stock if the expected return is positive:

$$\mathbb{E}[R] = p_V r_V + p_H r_H + p_L r_L > 0.$$

Figure 1.1 summarizes the structure.

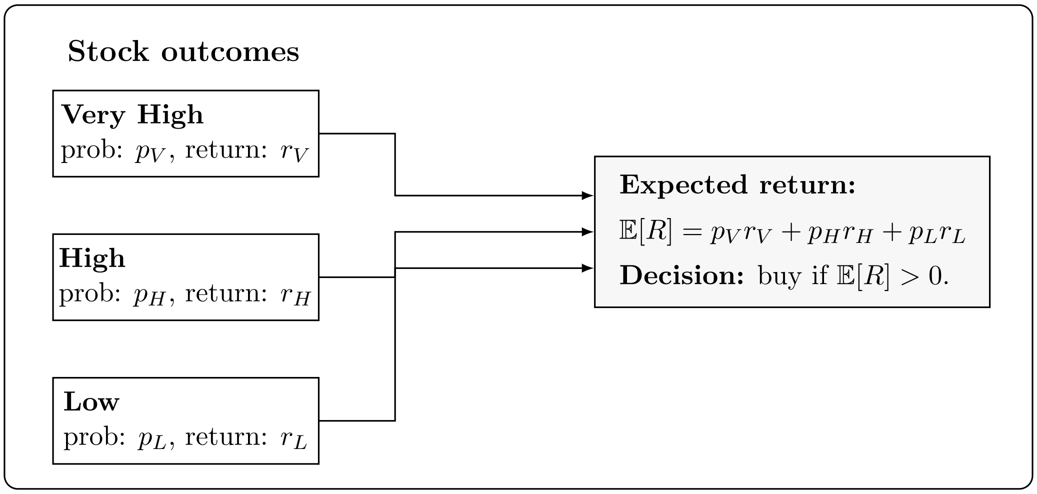

Figure 1.1: Three–tier model for tomorrow's price and the risk–neutral decision rule.

1.3 Example B: rain forecasting and incentives

Suppose a weather forecaster reports a number p (between 0 and 1) as her declared 'probability of rain tomorrow.' Let q be her *true* belief about the chance of rain.

The payment rule is as follows. If it rains tomorrow, she receives p. Otherwise, she keeps the initial p, but must pay $-\ln(1-p)$. So her payoff is:

- Rain: p
- No rain: $p + \ln(1-p)$

For an arbitrary true chance of rain $q \in (0,1)$, if the forecaster reports $p \in (0,1)$, her expected payoff is:

$$F(p) = q\,p + (1-q)\big(p + \ln(1-p)\big) = p + (1-q)\ln(1-p).$$

The forecaster chooses the report p that maximizes this quantity:

$$\hat{p} \;=\; \arg\max_{0<p<1} \; F(p).$$

Differentiate with respect to p:

$$F'(p) \;=\; 1 - \frac{1-q}{1-p}.$$

Setting $F'(p) = 0$ gives

$$1 - \frac{1-q}{1-p} = 0 \quad\Longleftrightarrow\quad 1-p = 1-q \quad\Longleftrightarrow\quad \hat{p} = q.$$

Since

$$F''(p) = -\frac{1-q}{(1-p)^2} < 0 \qquad (0 < p < 1),$$

the function $F(p)$ is strictly concave and $\hat{p} = q$ is the unique maximizer.

In words, for any fixed belief q, the forecaster maximizes her expected payoff by reporting $p = q$.

Figure 1.2 summarizes the structure.

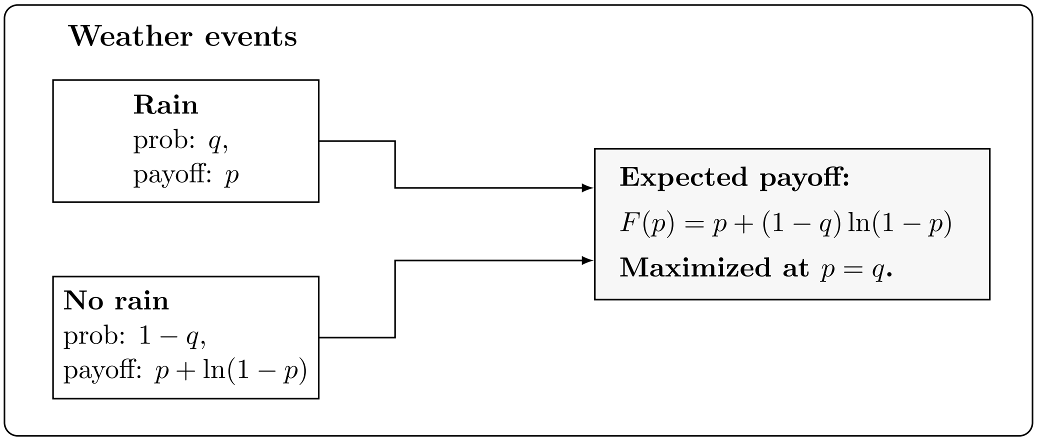

Figure 1.2: Rain forecasting diagram under the payoff rule rain $\to p$ and no rain $\to p + \ln(1-p)$.

1.4 Modeling checklist

1. Outcomes and events: identify what can happen.
2. Probabilities: quantify beliefs or frequencies for these outcomes.
3. Actions and payoffs: list the decisions and their consequences.
4. Decision rule: choose the action maximizing expected value.

KEEN 3Cs

Curiosity

Explore how decisions change if outcome categories are refined or merged. Identify what additional information reduces uncertainty.

Connections

Connect expected value reasoning here to the Law of Total Probability, Bayes theorem, and reliability approximations using bounds introduced later.

Creating Value

Propose a payoff or scoring rule that encourages honest weather forecasts for campus operations planning outdoor events.

Practice

A. Remember and Understand

1. [Remember]List actions, outcomes, probabilities, and returns in the stock example.
2. [Understand]State the payoff under the forecasting rule if it rains and if it does not rain.
3. [Understand]Compute $\mathbb{E}[R]$ in the stock example with $(3, 1, -1)$ and $(0.2, 0.5, 0.3)$.

B. Apply and Analyze

1. [Apply]Compute $\mathbb{E}[R]$ for $(4, 1, -2)$ and $(0.2, 0.6, 0.2)$ and decide.
2. [Apply]Compute the rain payoff p and no-rain payoff $p + \ln(1-p)$ for $q = 0.3$ and $p = 0.4$.
3. [Analyze]For $F(p) = p + (1-q)\ln(1-p)$ compute $F'(p)$ and verify $F'(q) = 0$.

C. Evaluate and Create

1. [Evaluate]Explain why reporting p very close to 0 or 1 may be risky under this rule if q is moderate.
2. [Create]Propose a threshold-based decision rule (e.g., cancel an event if p exceeds a threshold) and justify it.

Chapter 2

Events, Axioms, and Probability Models

Topics to Cover

- Sample spaces and events
- Event operations and diagrams
- Probability models and Kolmogorov's axioms
- Consequences: complement rule, addition rule, inclusion–exclusion, union bound
- σ-algebras as collections of allowable events

Learning Objectives

- Describe a sample space and event structure for simple systems.
- State the three probability axioms in plain language and formal notation.
- Manipulate events using complements, unions, intersections, and disjointness.
- Derive basic identities such as $P(A^c) = 1 - P(A)$ and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Explain what a σ-algebra is and why probability models are built on them.

2.1 Sample spaces and events

Before assigning probabilities, we must specify what can happen. A **sample space** Ω is the set of all possible outcomes. Examples:

- Rolling a die: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- Weather tomorrow: $\Omega = \{\text{rain}, \text{no rain}\}$.
- Packet arrival in a time slot: $\Omega = \{0, 1\}$ (no arrival or arrival).

An **event** is a subset of Ω. For the die example:

$$A = \{2, 4, 6\} \qquad \text{('even number')}.$$

Events are the objects to which we will assign probabilities.

2.2 Event operations

Given events A and B:

- A^c: complement—outcomes where A does not occur.
- $A \cup B$: union—outcomes where at least one occurs.
- $A \cap B$: intersection—outcomes where both occur.
- A and B are **disjoint** if $A \cap B = \emptyset$.

Venn diagrams

The following simple TikZ diagrams illustrate these operations.

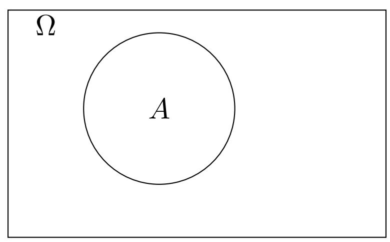

Figure 2.1: Event A inside sample space Ω.

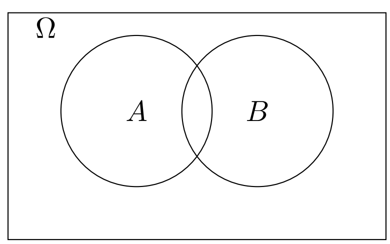

Figure 2.2: Events A and B and their overlap.

2.3 Probability models and axioms

A **probability model** consists of:

- a sample space Ω,
- a collection of allowable events $\mathcal{F}$,
- a probability assignment $P(\cdot)$.

2.3.1 The need for a collection of events

Not every subset of Ω needs to be (or should be) an event. In simple systems $\mathcal{F}$ includes all subsets, but in richer systems we only allow events we can describe or reason about.

A **σ-algebra** $\mathcal{F}$ is a collection of subsets of Ω satisfying:

1. $\Omega \in \mathcal{F}$.
2. If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$.
3. If $A_1, A_2, \dots \in \mathcal{F}$ then $\bigcup_{n\geq 1} A_n \in \mathcal{F}$.

This guarantees that the usual event operations stay within the model.

2.3.2 Kolmogorov's axioms

A probability measure $P : \mathcal{F} \to [0, 1]$ satisfies:

1. $P(A) \geq 0$ for every $A \in \mathcal{F}$.
2. $P(\Omega) = 1$.
3. For disjoint $A_1, A_2, \dots,$
$$P\left(\bigcup_{n\geq 1} A_n\right) = \sum_{n\geq 1} P(A_n).$$

2.4 Consequences of the axioms

Complement rule.
$$P(A^c) = 1 - P(A).$$

Monotonicity. If $A \subseteq B$ then $P(A) \leq P(B)$.

Addition rule (two events).

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Union bound.
$$P(A \cup B) \leq P(A) + P(B).$$

This will reappear in reliability calculations.

2.5 Examples

2.5.1 Two sensors

A system has two sensors that can fail. Let:

$$A = \{\text{sensor 1 fails}\}, \qquad B = \{\text{sensor 2 fails}\}.$$

If $P(A) = 0.1$, $P(B) = 0.1$, and they fail independently, then

$$P(A \cup B) = 0.1 + 0.1 - 0.1 \cdot 0.1 = 0.19.$$

If independence is not known:

$$P(A \cup B) \leq 0.1 + 0.1 = 0.2.$$

2.5.2 Weather

Let:

$$A = \{\text{rain}\}, \qquad A^c = \{\text{no rain}\}.$$

Then $P(A^c) = 1 - P(A)$.

When the forecaster reports a number p, the model in Chapter 1 describes how payoffs depend on events A and A^c.

KEEN 3Cs

Curiosity

What hidden assumptions lie behind defining a sample space? How might different teams model the same situation differently?

Connections

Event operations introduced here connect directly to the Law of Total Probability, Bayes theorem, and reliability analysis.

Creating Value

Define events clearly in an engineering scenario (sensing, networking, forecasting) so that decisions based on them reliably create value.

Practice

A. Remember and Understand

1. [Remember]Define sample space, event, and σ-algebra in your own words.
2. [Understand]Use the axioms to derive $P(A^c) = 1 - P(A)$.
3. [Understand]Draw a Venn diagram showing $A \cup B$, $A \cap B$, and A^c.

B. Apply and Analyze

1. [Apply]Compute $P(A \cup B)$ when $P(A) = 0.3$, $P(B) = 0.4$, and $P(A \cap B) = 0.1$.
2. [Analyze]A system fails if either component 1 or 2 fails. If $P(\text{fail}_1) = 0.1$, $P(\text{fail}_2) = 0.1$, give upper and lower bounds on system failure.
3. [Analyze]Let $A \subseteq B$. Show directly from the axioms that $P(A) \leq P(B)$.

C. Evaluate and Create

1. [Evaluate]For a sensing system, propose a sample space and three events relevant to an operations team.

2. [Create]Formulate a σ-algebra of meaningful events for a simple weather-based decision (e.g., cancel, delay, proceed).

Chapter 3

Counting Principles and Equally Likely Outcomes

Topics to Cover

- Product rule (rule of product) for multi-step processes
- Permutations and combinations; $\binom{n}{k}$ and factorial notation
- Equally likely outcome models and basic probabilities
- Simple counting trees and sanity checks

Learning Objectives

- Decompose multi-step experiments and apply the product rule to count outcomes.
- Compute permutations and combinations; use $\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$.
- Model basic systems with equally likely outcomes and compute probabilities by counting.
- Use small trees to check that counts match the process definition.

3.1 Product rule: multi-step counting

When an experiment proceeds in steps, and the number of options at each step is fixed (or can be tracked), the **product rule** gives the total number of outcomes as the product of the number of options at each step.

Example (two-step code). A device generates a two-symbol code: the first symbol is chosen from $\{A, B, C\}$ and the second from $\{0, 1\}$. There are $3 \times 2 = 6$ outcomes:

$$\{A0, A1, B0, B1, C0, C1\}.$$

3.2 Permutations and combinations

Permutation of n distinct items. Arrangements (order matters): $n! = n \cdot (n-1) \cdots 2 \cdot 1$.

k-permutations from n distinct items. Arrangements of k out of n (order matters):

$$P(n,k) = n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

Combinations of n choose k (order does not matter).

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Example (ID badges). Four distinct letters $\{A, B, C, D\}$; form a two-letter badge:

- If order matters (e.g., printed left-to-right), $P(4,2) = 4 \cdot 3 = 12$.
- If order does not matter (e.g., team of two), $\binom{4}{2} = 6$.

3.3 Equally likely outcome models

If a finite sample space Ω has N outcomes and each has probability $1/N$, then for any event $A \subseteq \Omega$,

$$P(A) = \frac{|A|}{|\Omega|} = \frac{|A|}{N}.$$

Example (cards). From a standard deck (ignore suits if needed), probability of drawing a specific rank among 13 equally likely ranks is $1/13$.

Example (bitstrings). Choose a length-n bitstring uniformly from $\{0,1\}^n$. The number with exactly k ones is $\binom{n}{k}$, so

$$P(\text{exactly } k \text{ ones}) = \frac{\binom{n}{k}}{2^n}.$$

3.4 Sanity checks using counting trees

Trees are useful as quick structural checks that the product rule and $\binom{n}{k}$ counts match the process.

3.5 Worked examples

Example (teams from a class). From n students, how many distinct teams of k can be formed? Order does not matter:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

If roles are distinct (e.g., captain and co-captain), then order matters:

$$P(n,2) = n(n-1).$$

Example (quality control). A batch has n items; d defective, $n-d$ good. If two are chosen at random without replacement, the probability that both are good is

$$\frac{\binom{n-d}{2}}{\binom{n}{2}}.$$

KEEN 3Cs

Curiosity

Where might an "equally likely" assumption fail? Propose one variation of a counting model that invalidates symmetry.

Connections

Connect counting to later topics: binomial and negative binomial models, Poisson approximations, and reliability compositions that rely on combinatorial structure.

Creating Value

Design a simple sampling or inspection plan for a real decision (e.g., quality control or scheduling), and explain how counting supports a defensible choice.

Practice

A. Remember and Understand

1. [Remember]State the product rule and give one short example.
2. [Understand]Explain the difference between $P(n,k)$ and $\binom{n}{k}$ with a concrete scenario.
3. [Understand]In the equally likely model on a finite Ω, explain why $P(A) = |A|/|\Omega|$.

B. Apply and Analyze

1. [Apply]A code has 3 letters (A, B, C) followed by 2 digits (0–9). Count how many codes if letters can repeat and digits can repeat.
2. [Apply]From 10 students, how many ordered pairs (captain, co-captain) can be formed? How many unordered pairs?
3. [Analyze]A length-8 bitstring is chosen uniformly. Compute $P(\text{exactly 3 ones})$ and $P(\text{at least 3 ones})$.

C. Evaluate and Create

1. [Evaluate]In a simple hiring process with two rounds (screen, interview), propose a counting model that matches capacity constraints and justify it.
2. [Create]Create a three-step tree for a process of your choice and use it to compute a count both from the tree and from a closed-form expression; explain the match.

Chapter 4

Conditional Probability and the Law of Total Probability

Topics to Cover

- Conditional probability and its interpretation
- Multiplication rule for joint events
- Partition of the sample space
- Law of Total Probability
- Conditioning as information update

Learning Objectives

- Define and compute conditional probabilities.
- Derive the multiplication rule for probabilities.
- Use partitions to decompose complex probability calculations.
- Apply the Law of Total Probability in structured settings.
- Interpret conditioning as updating beliefs given new information.

4.1 Motivation: Updating with information

In earlier chapters, probabilities were assigned to events before observing additional information. In many engineering settings, information arrives sequentially.

Examples:

- A sensor reports a signal.
- A test result becomes available.

- A partial observation is revealed.

We now formalize how probabilities change when new information is known.

4.2 Conditional probability

Definition. Let A and B be events with $P(B) > 0$. The **conditional probability** of A given B is:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Interpretation.

- We restrict attention to outcomes in B
- We renormalize probabilities within B

Key identity.

$$P(A \cap B) = P(A \mid B)\, P(B).$$

This is called the **multiplication rule**.

4.3 Examples of conditional probability

4.3.1 Card example

Draw one card from a standard deck.

Let:

$$A = \{\text{card is a heart}\}, \quad B = \{\text{card is red}\}.$$

Then:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{13/52}{26/52} = \frac{1}{2}.$$

4.3.2 System reliability

Let:

$$A = \{\text{component 1 works}\}, \quad B = \{\text{component 2 works}\}.$$

Then:

$$P(A \mid B)$$

represents the reliability of component 1 given component 2 is functioning.

4.4 Multiplication rule and sequential structure

The multiplication rule generalizes:

$$P(A \cap B \cap C) = P(A \mid B \cap C)\, P(B \mid C)\, P(C).$$

This rule allows us to compute probabilities of multi-stage processes.

Example (two-step experiment).

$$P(A \cap B) = P(A)\, P(B \mid A).$$

4.5 Partitions of the sample space

A collection of events $\{B_1, B_2, \ldots, B_n\}$ is a **partition** if:

- $B_i \cap B_j = \emptyset$ for $i \neq j$
- $\bigcup_{i=1}^{n} B_i = \Omega$

Partitions represent mutually exclusive, exhaustive scenarios.

4.6 Law of Total Probability

Let $\{B_1, B_2, \ldots, B_n\}$ be a partition with $P(B_i) > 0$. Then for any event A:

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)\, P(B_i).$$

Interpretation. We compute $P(A)$ by:

- conditioning on which scenario B_i occurs
- weighting by $P(B_i)$

4.7 Examples of the Law of Total Probability

4.7.1 Manufacturing example

A factory has two machines:

- Machine 1 produces 70% of items (event B_1)
- Machine 2 produces 30% (event B_2)

Let:

$$A = \{\text{item is defective}\}.$$

Given:

$$P(A \mid B_1) = 0.01, \quad P(A \mid B_2) = 0.03.$$

Then:

$$P(A) = 0.01(0.7) + 0.03(0.3) = 0.016.$$

4.7.2 Weather example

Let:

$$B_1 = \{\text{cloudy}\}, \quad B_2 = \{\text{clear}\}.$$

Then any probability of rain can be decomposed as:

$$P(\text{rain}) = P(\text{rain} \mid \text{cloudy})P(\text{cloudy}) + P(\text{rain} \mid \text{clear})P(\text{clear}).$$

4.8 Conditioning as filtering

Conditional probability can be viewed as filtering the sample space:

- Original space: Ω
- After observing B: new space is B

Within B, probabilities are rescaled to sum to 1.

4.9 Modeling checklist

1. Identify events A, B
2. Check that $P(B) > 0$
3. Compute $P(A \cap B)$
4. Use:
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$
5. If multiple cases exist, define a partition
6. Apply:
$$P(A) = \sum_i P(A \mid B_i)P(B_i)$$

KEEN 3Cs

Curiosity

How does additional information change decision outcomes? Explore how conditioning alters probabilities in real systems.

Connections

Conditional probability connects to:

- Bayes' theorem (next chapter)
- Independence
- Random variables and distributions

Creating Value

Design a decision rule for a system where information arrives sequentially (e.g., sensor networks, testing pipelines) and justify how probabilities are updated.

Practice

A. Remember and Understand

1. [Remember]Define conditional probability.
2. [Understand]State the multiplication rule.
3. [Understand]Define a partition of a sample space.

B. Apply and Analyze

1. [Apply]A card is red. What is the probability it is a heart?
2. [Apply]Compute $P(A)$ using the Law of Total Probability given a two-case partition.
3. [Analyze]Show that $P(A \cap B) = P(A \mid B)P(B)$.

C. Evaluate and Create

1. [Evaluate]Explain how conditioning changes a probability model.
2. [Create]Construct a partition-based model for a reliability system with two modes of operation.

Chapter 5

Bayes' Theorem

Topics to Cover

- Bayes' theorem
- Prior and posterior probabilities
- Use of partitions in probabilistic inference

Learning Objectives

- Derive Bayes' theorem from conditional probability.
- Compute posterior probabilities using partitions.
- Interpret Bayes' rule as updating beliefs after observing an event.

5.1 Motivation: reversing conditioning

From Chapter 4, we compute:

$$P(A \mid B)$$

In many situations:

- an event A is observed
- we want to infer which underlying event E_i led to this observation

This motivates reversing conditional probabilities.

5.2 Bayes' Theorem

Starting point. From conditional probability:

$$P(E \cap A) = P(E \mid A)P(A)$$

$$P(E \cap A) = P(A \mid E)P(E)$$

Theorem (Bayes). For $P(A) > 0$:

$$\boxed{P(E \mid A) = \frac{P(A \mid E)P(E)}{P(A)}}$$

5.3 Role of partitions

Let $\{E_1, E_2, \ldots, E_n\}$ be a partition of Ω:

- $E_i \cap E_j = \emptyset \quad (i \neq j)$
- $\bigcup_{i=1}^{n} E_i = \Omega$

Then any event A can be written as:

$$A = \bigcup_{i=1}^{n} (A \cap E_i)$$

Since the terms are disjoint:

$$P(A) = \sum_{i=1}^{n} P(A \cap E_i)$$

Using conditional probability:

$$P(A) = \sum_{i=1}^{n} P(A \mid E_i)\, P(E_i)$$

This is the Law of Total Probability.

5.4 Bayes' rule with partitions

Let $\{E_1, \ldots, E_n\}$ be a partition.

Then:

$$\boxed{P(E_i \mid A) = \frac{P(A \mid E_i)\, P(E_i)}{\sum_{j=1}^{n} P(A \mid E_j)\, P(E_j)}}$$

5.5 Diagram: decomposition using partitions

$$P(A) = \sum_{i} P(A \cap E_i) = \sum_{i} P(A \mid E_i)\, P(E_i)$$

5.6 Example: two possible cases

Suppose:

- exactly one of E_1 or E_2 occurs
- event A is observed

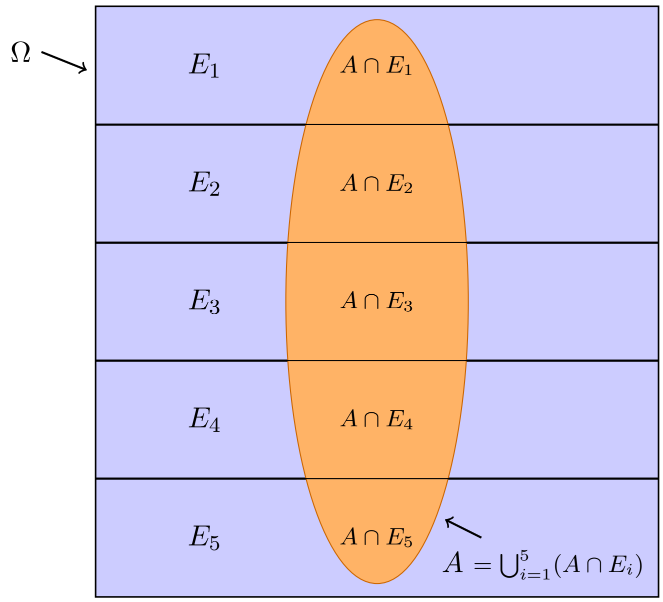

Figure 5.1: Decomposition $A = \bigcup_{i=1}^{5}(A \cap E_i)$

Then:

$$P(E_1 \mid A) = \frac{P(A \mid E_1)P(E_1)}{P(A \mid E_1)P(E_1) + P(A \mid E_2)P(E_2)}$$

Interpretation. The observation A updates our belief about whether E_1 or E_2 occurred.

5.7 Example: coin selection

- One of two coins is selected
- E_1: coin 1 chosen, E_2: coin 2 chosen
- A: head observed

Then:

$$P(E_1 \mid A) = \frac{P(A \mid E_1)P(E_1)}{P(A \mid E_1)P(E_1) + P(A \mid E_2)P(E_2)}$$

5.8 Interpretation

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

- Prior: $P(E_i)$
- Likelihood: $P(A \mid E_i)$
- Posterior: $P(E_i \mid A)$

The denominator ensures normalization:

$$\sum_i P(E_i \mid A) = 1$$

5.9 Modeling checklist

1. Identify a partition $\{E_i\}$
2. Specify prior probabilities $P(E_i)$
3. Specify likelihoods $P(A \mid E_i)$
4. Compute:
$$P(A) = \sum_i P(A \mid E_i)\, P(E_i)$$
5. Compute posterior probabilities $P(E_i \mid A)$

KEEN 3Cs

Curiosity

How does observing an event change our beliefs about underlying causes? What happens when prior probabilities are very small?

Connections

Bayes' theorem builds directly on:

- Conditional probability (Chapter 4)
- Law of Total Probability

Creating Value

Construct a model where observations are used to infer which scenario among several possibilities occurred.

Practice

A. Remember and Understand

1. [Remember]State Bayes' theorem.
2. [Understand]Explain prior and posterior probabilities.

B. Apply and Analyze

1. [Apply]Compute $P(E_1 \mid A)$ for two cases.
2. [Analyze]Verify that the denominator equals $P(A)$.

C. Evaluate and Create

1. [Evaluate]Explain how prior probabilities influence the result.
2. [Create]Construct a three-case inference model.

Chapter 6

Independence of Events

Topics to Cover

- Definition of independence
- Independence via conditional probability
- Product rule for independent events
- Independence and complements
- Pairwise and mutual independence

Learning Objectives

- Define independence using conditional probability and joint probability.
- Verify independence using multiple equivalent conditions.
- Apply independence properties to complements and derived events.
- Distinguish between pairwise and mutual independence.

6.1 Definition of independence

Let A and B be events.

Definition (via conditioning). Assume $P(A) > 0$. B is independent of A if:

$$P(B \mid A) = P(B)$$

Similarly, if $P(B) > 0$, A is independent of B if:

$$P(A \mid B) = P(A)$$

Interpretation. Independence means that knowledge of one event does not change the probability of the other.

6.2 Equivalent formulation

Proposition. If $P(A) > 0$ and $P(B) > 0$, then:

$$P(B \mid A) = P(B) \iff P(A \cap B) = P(A)P(B)$$

Proof. From conditional probability:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

Thus:

$$P(B) = \frac{P(A \cap B)}{P(A)} \quad \Rightarrow \quad P(A \cap B) = P(A)P(B)$$

Conversely, if:

$$P(A \cap B) = P(A)P(B)$$

then:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = P(B)$$

6.3 Symmetry of independence

Proposition. If $P(A) > 0$ and $P(B) > 0$, then:

$$P(B \mid A) = P(B) \iff P(A \mid B) = P(A)$$

Proof. If:

$$P(A \cap B) = P(A)P(B)$$

then:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A)$$

Thus independence is symmetric.

6.4 Independence and complements

Proposition. If A and B are independent, then:

- A^c and B are independent
- A and B^c are independent
- A^c and B^c are independent

Proof. First, consider A^c and B:

$$P(A^c \cap B) = P(B) - P(A \cap B)$$

Using independence:

$$= P(B) - P(A)P(B) = P(B)(1 - P(A)) = P(B)P(A^c)$$

Thus:

$$P(A^c \cap B) = P(A^c)P(B)$$

which shows independence.

The other cases follow similarly.

6.5 Conditional probabilities under independence

Proposition. If A and B are independent and $P(A) > 0$:

$$P(B \mid A) = P(B)$$

If $P(A^c) > 0$:

$$P(B \mid A^c) = P(B)$$

Proof.

$$P(B \mid A^c) = \frac{P(B) - P(A \cap B)}{P(A^c)} = \frac{P(B) - P(A)P(B)}{1 - P(A)} = P(B)$$

Thus independence extends to complements.

6.6 Basic properties

Property. If A and B are independent:

$$P(A \cap B) = P(A)P(B)$$

Property. If A and B are independent:

$$P(A \mid B) = P(A), \quad P(B \mid A) = P(B)$$

6.7 Multiple events

6.7.1 Pairwise independence

Events A, B, and C are pairwise independent if:

$$P(A \cap B) = P(A)P(B)$$

$$P(B \cap C) = P(B)P(C)$$

$$P(A \cap C) = P(A)P(C)$$

6.7.2 Mutual independence

Events A, B, and C are mutually independent if:

- they are pairwise independent, and
- $$P(A \cap B \cap C) = P(A)P(B)P(C)$$

Important. Pairwise independence does not imply mutual independence.

6.8 Extension to n events

Events $A_1, A_2, \ldots, A_n$ are independent if for every subset:

$$P(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) = P(A_{i_1})P(A_{i_2}) \cdots P(A_{i_k})$$

for all $k \geq 2$.

6.9 Modeling checklist

1. Identify events A, B
2. Compute:
$$P(A), \quad P(B), \quad P(A \cap B)$$
3. Check:
$$P(A \cap B) = P(A)P(B)$$
4. Alternatively verify:
$$P(A \mid B) = P(A)$$

KEEN 3Cs

Curiosity

Can two events appear related but still be independent? Find examples where intuition is misleading.

Connections

Independence builds on:

- Conditional probability (Chapter 4)
- Bayes' theorem (Chapter 5)

Creating Value

Construct an example of three events that are pairwise independent but not mutually independent.

Practice

A. Remember and Understand

1. [Remember]Define independence.
2. [Understand]State the product rule for independent events.

B. Apply and Analyze

1. [Apply]Check whether given events are independent.
2. [Analyze]Verify independence for complements.

C. Evaluate and Create

1. [Evaluate]Explain the difference between pairwise and mutual independence.
2. [Create]Construct events that are pairwise but not mutually independent.

Chapter 7

Random Variables

Topics to Cover

- Motivation for random variables
- Definition as a mapping
- Domain, range, and codomain
- Direct and inverse images
- Discrete random variables
- Probability mass function (PMF)

Learning Objectives

- Understand why random variables are introduced.
- Define a random variable as a function.
- Interpret mappings from outcomes to numbers.
- Compute probabilities using inverse images.
- Construct a probability mass function.

7.1 Motivation: assigning numerical values

Consider a simple experiment: tossing a coin.

$$\Omega = \{H, T\}$$

Instead of working directly with outcomes, we assign a number to each outcome:

$$H \mapsto 1, \quad T \mapsto 0$$

This assignment allows us to study the experiment using numerical values.

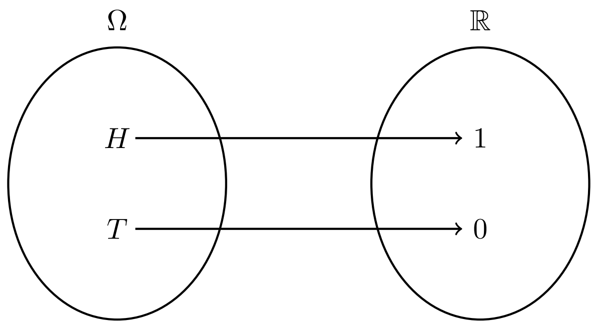

Figure 7.1: Mapping outcomes to numerical values

7.2 Definition of a random variable

Definition. A **random variable** is a function:

$$X : \Omega \to \mathbb{R}$$

Each outcome $\omega \in \Omega$ is mapped to a real number:

$$\omega \mapsto X(\omega)$$

7.3 Random variables as functions between sets

We view a random variable as a mapping between two sets.

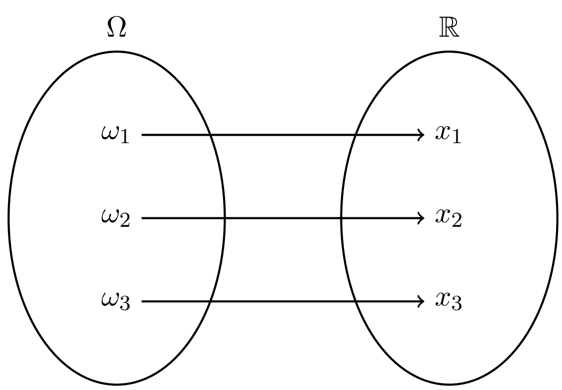

Figure 7.2: Random variable as a mapping from Ω to $\mathbb{R}$

7.4 Example: rolling a die

Consider the experiment of rolling a die:

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

Define a random variable:

$$X(\omega) = \omega$$

This assigns to each outcome its numerical value.

Another example. Define a different random variable:

$$X_2(\omega) = \begin{cases} 1 & \text{if } \omega \in \{4,5,6\} \\ 0 & \text{if } \omega \in \{1,2,3\} \end{cases}$$

This random variable indicates whether the outcome is large.

7.5 Visualization of a random variable on a die

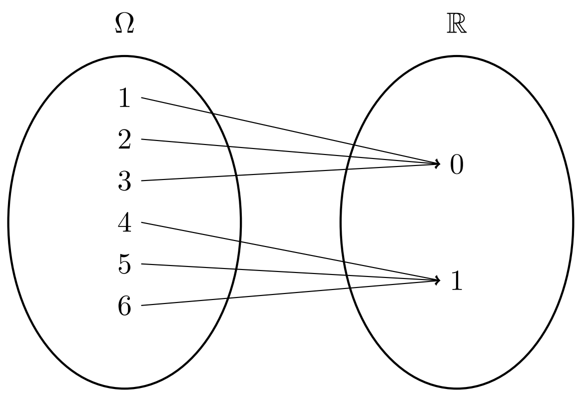

Figure 7.3: Indicator random variable: mapping outcomes to $\{0, 1\}$

7.6 Domain, codomain, and range

For a random variable:

$$X : \Omega \to \mathbb{R}$$

- Domain: the set Ω
- Codomain: the set $\mathbb{R}$
- Range: the set of values actually taken by X

Example. For:

$$X_2(\omega) = \begin{cases} 1 & \omega \in \{4,5,6\} \\ 0 & \text{otherwise} \end{cases}$$

the range is:

$$\{0, 1\}$$

7.7 Different random variables on the same experiment

Different random variables can be defined on the same sample space.

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

Examples:

$$X(\omega) = \omega$$

$$X_2(\omega) = \begin{cases} 1 & \omega \in \{4, 5, 6\} \\ 0 & \text{otherwise} \end{cases}$$

These represent different ways of extracting information from the same experiment.

7.8 Direct and inverse images

Let $A \subseteq \Omega$ and $B \subseteq \mathbb{R}$.

7.8.1 Direct image

The direct image of a subset $A \subseteq \Omega$ is:

$$X(A) = \{X(\omega) : \omega \in A\}$$

This gives the set of values taken by X on A.

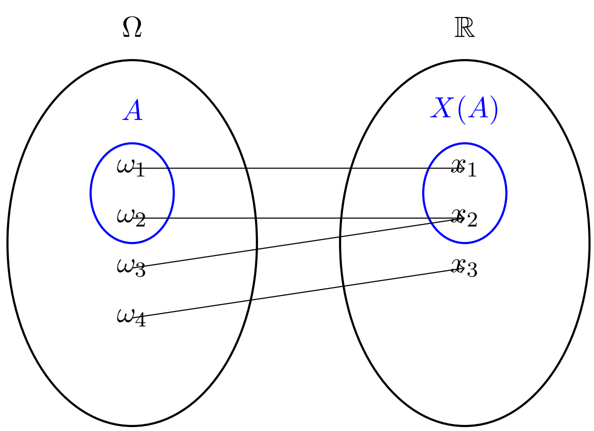

Figure 7.4: Direct image: $A = \{\omega_1, \omega_2\}$ maps to $X(A) = \{x_1, x_2\}$

7.8.2 Inverse image

The inverse image of a subset $B \subseteq \mathbb{R}$ is:

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}$$

This gives the set of outcomes that map into B.

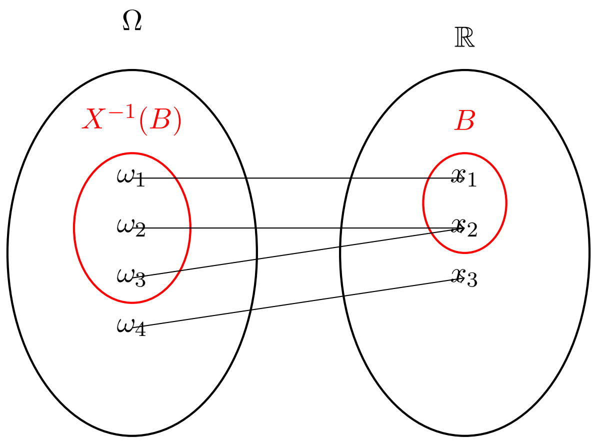

Figure 7.5: Inverse image: $B = \{x_1, x_2\}$ yields $X^{-1}(B) = \{\omega_1, \omega_2, \omega_3\}$

7.9 Probabilities using inverse images

For any set $B \subseteq \mathbb{R}$:

$$P(X \in B) = P\big(X^{-1}(B)\big)$$

7.10 Example: coin toss

$$\Omega = \{H, T\}, \quad X(H) = 1, \quad X(T) = 0$$

$$P(X = 1) = P(\{\omega : X(\omega) = 1\}) = P(\{H\})$$

$$P(X \in \{0, 1\}) = P(\{H, T\}) = 1$$

7.11 Discrete random variables

Definition. A random variable X is called **discrete** if it takes values in a finite or countable set:

$$\{x_1, x_2, x_3, \dots\}$$

7.12 Probability mass function (PMF)

Definition. The probability mass function (PMF) of X is defined as:

$$p_X(x) = P(X = x)$$

Using inverse images.

$$p_X(x) = P(\{\omega : X(\omega) = x\})$$

7.13 Example: number of heads

Consider two coin tosses:

$$\Omega = \{HH, HT, TH, TT\}$$

Define:

$$X = \text{number of heads}$$

$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0$$

Thus:

$$X \in \{0, 1, 2\}$$

$$p_X(0) = P(\{TT\}) = \frac{1}{4}$$

$$p_X(1) = P(\{HT, TH\}) = \frac{1}{2}$$

$$p_X(2) = P(\{HH\}) = \frac{1}{4}$$

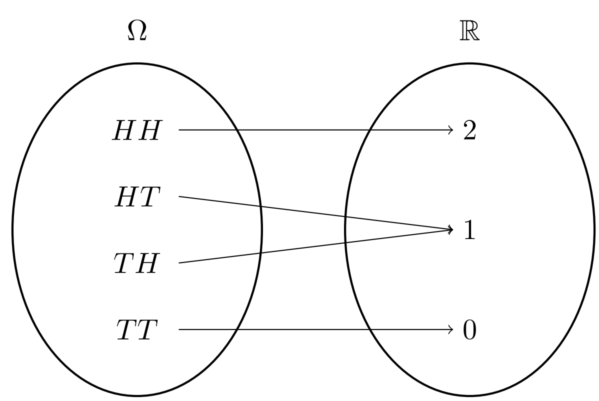

Figure 7.6: Random variable: number of heads

7.14 Properties of the PMF

- $p_X(x) \geq 0$
- $\sum_x p_X(x) = 1$

7.15 Modeling checklist

1. Define the sample space Ω
2. Define the random variable $X : \Omega \to \mathbb{R}$

3. Identify the range of X
4. Compute inverse images:
$$X^{-1}(x)$$
5. Compute:
$$p_X(x) = P(X = x)$$

KEEN 3Cs

Curiosity

How does mapping outcomes to numbers simplify probability problems?

Connections

Random variables connect sample spaces to numerical representations of uncertainty.

Creating Value

Define a random variable for an experiment and compute its PMF.

Practice

A. Remember and Understand

1. [Remember]Define a discrete random variable.
2. [Understand]Define the PMF.

B. Apply and Analyze

1. [Apply]Compute a PMF from a given experiment.
2. [Analyze]Use inverse images to compute probabilities.

C. Evaluate and Create

1. [Evaluate]Compare two different random variables on the same space.
2. [Create]Construct a PMF for a real system.

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